

SRUTHI MANGALA SURESH

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PROFESSIONAL SUMMARY

Master's graduate in Electrical Engineering (Smart Systems) with hands-on experience building Generative AI and Retrieval-Augmented Generation (RAG) systems, and a software development background at Mercedes-Benz and Bosch. Skilled in Python, LLM evaluation, and containerized deployment, with a focus on turning applied AI research into working, well-engineered software.

SKILLS AND LANGUAGES

- **Programming:** Python, C, SQL
- **AI Tools & Frameworks:** Weaviate, Ollama, Hugging Face, FastAPI, PostgreSQL
- **Cloud & MLOps:** AWS, Amazon Bedrock, Amazon SageMaker, Terraform
- **DevOps:** Docker, Docker Compose, Kubernetes, Git, GitLab CI/CD
- **Tools:** Atlassian Jira, Confluence
- **Languages:** English (C1), German (B2.1)

PROFESSIONAL EXPERIENCE

University of Stuttgart, Stuttgart, Germany

Feb 2026 – Sep 2026

Graduate Student Assistant - Part Time

Warehouse Automation & Task Scheduling Optimization

- Modelled warehouse pick-and-place operations in Siemens Plant Simulation to evaluate throughput and scheduling efficiency under varying layouts and demand.
- Implemented path planning algorithms to establish baseline routing and scheduling performance.
- Extended the simulation to a deep RL framework for adaptive task scheduling.

Vector Consulting Services GmbH and Universität Stuttgart, Stuttgart, Germany

Jan 2026 – Jun 2026

Master Thesis Student

Generative AI Support for Test and Traceability

- Designed and built a RAG-based platform for automotive requirements traceability and test management, using Weaviate vector search, BGE-M3 embeddings, and a FastAPI and PostgreSQL backend orchestrated with Docker Compose.
- Implemented an LLM-as-judge evaluation pipeline benchmarking multiple language models on accuracy, output quality, and hardware/scalability trade-offs, using multi-label classification metrics.
- Ran local LLMs via Ollama under constrained GPU memory, tuning chat and embedding model configurations for performance.

Mercedes-Benz AG, Sindelfingen, Germany

Apr 2025 – Oct 2025

Internship- Full Time

Software Developer Intern

- Built and maintained GitLab CI/CD pipelines to automate testing and deployment for an internal engineering tool used across various teams.
- Improved the tool's documentation and usability, reducing onboarding friction for cross-team users.

Mercedes-Benz AG, Sindelfingen, Deutschland

Aug 2024 – Mar 2025

Working Student - Part Time

Software Development and Automation for ADAS projects

- Developed a Python and Jinja-based automation tool that generated structured Confluence documentation, reducing manual documentation effort for ADAS projects.

Robert Bosch GmbH, Feuerbach, Germany

Dec 2023 – May 2024

Working Student - Part Time

Fault Diagnostics and State Management for ADAS

- Enhanced a MATLAB-based ADAS state-management tool, improving fault-detection efficiency and diagnostic coverage.

University of Stuttgart, Stuttgart, Germany

Jul 2023 – Mar 2025

Student Assistant - Part Time

Python application development for university admissions data management

- Developed and tested Python GUI applications to streamline admissions data acquisition and sorting.
- Designed and maintained university web content using OpenCMS.

Deloitte USI, Bengaluru, India

Business Technology Analyst - Full Time

Jan 2022 – Feb 2023

Medicaid for State Government, USA

- Designed and executed functional and regression test cases for a Medicaid web application serving a U.S. state government.

PROJECTS

Research Thesis at Universität Stuttgart

Oct 2024 – Mar 2025

Framework for Static Function Offloading in Distributed Vehicular Cloud/Edge Networks

- Developed a framework for dynamic computation offloading in distributed edge-cloud systems
- Built a prototype using optimization algorithms, Docker, and Kubernetes.

EDUCATION

Universität Stuttgart, Stuttgart, Germany

Sep 2026

Master of Science in Electrical Engineering with a major in Smart Systems

Grade 1,7

Relevant Coursework : Deep Learning, Software Engineering, Industrial Automation Systems, Automotive RADAR, Embedded Systems, Detection and Pattern Recognition

APJ Abdul Kalam Technological University, Thiruvananthapuram, India

Sep 2021

Bachelor of Technology (Honors) in Electrical and Electronics Engineering

Grade 1,4

Relevant Coursework: Object Oriented Programming, C Programming, Soft Computing, Microprocessors and Embedded Systems, Engineering Mathematics

PUBLICATIONS

[YOLOv5 based Open-Source UAV for Human Detection during Search And Rescue \(SAR\)](#)

2022

Developed an open-source YOLOv5-based UAV for human detection in flood scenarios, enabling rapid rescue response.

SCHOLARSHIPS

Bosch ME-Stipendium

Oct 2024 – Mar 2025

Robert Bosch GmbH, Reutlingen, Germany

Scholarship for students studying Electrical Engineering who have a good academic track record